

Worksheet 4 Solution

Problem 1: $x^2 y'' - 3xy' + 10y = 0$

Assume $y = x^m$, $y' = mx^{m-1}$, $y'' = m(m-1)x^{m-2}$

Substituting into the ODE, we get

$$x^2(m(m-1)x^{m-2}) - 3x(mx^{m-1}) + 10x^m = 0$$

$$m(m-1) - 3m + 10 = 0$$

$$m^2 - 4m + 10 = 0 \Rightarrow m = 2 \pm i\sqrt{6}$$

$$y = x^2 (C_1 \cos(\sqrt{6} \log x) + C_2 \sin(\sqrt{6} \log x))$$

Problem 2: $x^2 y'' + xy' + 9y = 0$

Assume $y = x^m$, $y' = mx^{m-1}$, $y'' = m(m-1)x^{m-2}$

Substituting into the ODE, we get

$$x^2 m(m-1)x^{m-2} + xmx^{m-1} + 9x^m = 0$$

$$\Rightarrow m(m-1) + m + 9 = 0$$

$$m^2 + 9 = 0 \Rightarrow m = \pm 3i$$

$$y = C_1 \cos(3 \log(x)) + C_2 \sin(3 \log(x))$$

Given that $y(1) = 0$, $y'(1) = 2.5$

$$y(1) = C_1 = 0$$

$$y'(x) = -C_1 \sin(3 \log x) \frac{3}{x} + C_2 \cos(3 \log x) \cdot \frac{3}{x}$$

$$y'(1) = 3C_2 = 2.5 \Rightarrow C_2 = 0.834$$

Hence the solution is

$$y(x) = 0.834 \sin(3 \log x)$$

Problem 3:

$$(a) \quad y'' + y' + \left(\pi^2 + \frac{1}{4}\right)y = e^{-x/2} \sin \pi x$$

Solution of the homogenous problem:

$$\lambda^2 + \lambda + \pi^2 + \frac{1}{4} = 0$$

$$\Rightarrow \lambda = \frac{-1 \pm \sqrt{1 - 4\pi^2 - 1}}{2} = \frac{-1}{2} \pm i\pi$$

$$y_h = e^{-x/2} (C_1 \cos \pi x + C_2 \sin \pi x)$$

$$y_h = C_1 e^{-x/2} \cos \pi x + C_2 e^{-x/2} \sin \pi x$$

We try a particular solution of the form

$$y_p = A x e^{-x/2} \cos(\pi x) + B x e^{-x/2} \sin(\pi x)$$

$$y_p' = \frac{1}{2} e^{-x/2} [\cos(\pi x)(2\pi Bx - A(x-2)) - \sin(\pi x)(2\pi Ax + B(x-2))]$$

$$y_p'' = \frac{1}{4} e^{-x/2} [\sin(\pi x)(4\pi A(x-2) + B(-4\pi^2 x + x - 4)) + \cos(\pi x)(A(-4\pi^2 x + x - 4) - 4\pi B(x-2))]$$

We now substitute in the original equation,

$$y'' + y' + \left(\pi^2 + \frac{1}{4}\right)y = 2\pi e^{-x/2} \left(B \cos \pi x - A \sin(\pi x)\right) \\ = e^{-x/2} (\sin \pi x)$$

from where $B=0$ and $A = -\frac{1}{2\pi}$

So, $y_p = -\frac{1}{2\pi} x e^{-x/2} \cos(\pi x)$

The general solution is

$$y = y_h + y_p = e^{-x/2} \left(\left(C_1 - \frac{x}{2\pi} \right) \cos \pi x + C_2 \sin \pi x \right)$$

(b) $8y'' - 6y' + y = 6 \cosh x$

Solution of the homogenous problem

$$8\lambda^2 - 6\lambda + 1 = 0 \Rightarrow \lambda = \frac{1}{4}, \frac{1}{2}$$

$$y_h = C_1 e^{x/2} + C_2 e^{x/4}$$

We try a particular solution of the form

$$y_p = A \cosh x + B \sinh x$$

$$y_p' = A \sinh x + B \cosh x$$

$$y_p'' = A \cosh x + B \sinh x$$

Substituting into the differential equation
 $(9A - 6B) \cosh x + (9B - 6A) \sinh x = 6 \cosh x$
 $\Rightarrow$

$$9A - 6B = 6 \quad ; \quad 9B - 6A = 0$$

$$\Rightarrow A = \frac{6}{5}, \quad B = \frac{4}{5}$$

General solution $\rightarrow$

$$y = c_1 e^{x/2} + c_2 e^{x/4} + \frac{6}{5} \cosh x + \frac{4}{5} \sinh x$$

Given that $y(0) = 0.2$, $y'(0) = 0.05$

$$y(0) = c_1 + c_2 + \frac{6}{5} = 0.2$$

$$y'(0) = \frac{1}{2} c_1 + \frac{1}{4} c_2 + \frac{4}{5} = 0.05$$

$$\Rightarrow c_1 = -2, \quad c_2 = 1$$

$$y = -2 e^{x/2} + e^{x/4} + \frac{6}{5} \cosh x + \frac{4}{5} \sinh x$$

Problem 4: $y'' + y = \cos \omega t$

solution of the homogeneous problem $\rightarrow$

$$\lambda^2 + 1 = 0 \quad \lambda = \pm i$$

$$y_h = c_1 \cos t + c_2 \sin t$$

let $y_p = A \cos \omega t + B \sin \omega t$

$$y_p' = -\omega A \sin \omega t + \omega B \cos \omega t$$

$$y_p'' = -\omega^2 A \cos \omega t - \omega^2 B \sin \omega t$$

Substituting into the differential equation

$$A(1 - \omega^2) \cos \omega t + B(1 - \omega^2) \sin \omega t = \cos \omega t$$

$$\Rightarrow A = \frac{1}{1 - \omega^2} ; B = 0$$

So, the general solution is

$$y = c_1 \cos t + c_2 \sin t + \frac{1}{1 - \omega^2} \cos \omega t$$

Problem 5 : To model the circuit we sum the drops of voltage along the RC loop

$$E(t) - IR - \frac{Q}{C} = 0$$

$$E(t) - IR - \frac{1}{C} \int I(t) dt = 0$$

Differentiating, $E'(t) - I'R - \frac{1}{C} I = 0$

$$I' + \frac{1}{RC} I = \frac{E'}{R}$$

If E is constant, then $E(t) = E_0$, then $E' = 0$

$$I' + \frac{1}{RC} I = 0$$

Solution is $I(t) = A e^{-\frac{1}{RC} t}$

at $t=0$, we have $I(0)$, then

$$A = I(0)$$

$$I(t) = I(0) e^{-\frac{1}{RC} t}$$

(b) Given that $E = E_0 \sin \omega t$, we have

$$I' + \frac{1}{RC} I = \frac{E_0 \omega \cos \omega t}{R}$$

$$I(t) e^{\frac{t}{RC}} = \int \frac{E_0 \omega \cos \omega t}{R} \cdot e^{\frac{t}{RC}} dt + C_1$$

$$I(t) e^{\frac{t}{RC}} = \frac{E_0 \omega}{R} \frac{e^{\frac{t}{RC}}}{\left(\frac{1}{R^2 C^2} + \omega^2\right)} \left[\frac{1}{RC} \cos \omega t + \omega \sin \omega t \right] + C_1$$

$$I(t) = C_1 e^{-\frac{t}{RC}} + \frac{\omega E_0 C}{1 + (\omega RC)^2} (\cos \omega t + \omega RC \sin \omega t)$$

Problem 6. The given differential equation

$$y'' + R(x)y' + S(x)y = f(x)$$

Now using variation of parameters,

$$y_1(x) = f(x), \quad y_2(x) = xf(x)$$

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} f(x) & xf(x) \\ f'(x) & f(x) + xf'(x) \end{vmatrix}$$

$$\Rightarrow W(y_1, y_2) = [f^2(x) + xf'(x)f(x) - xf(x)f'(x)] = f^2(x)$$

$$y_p(x) = Ay_1 + By_2$$

where

$$A = - \int \frac{y_2 R}{W} dx, \quad B = \int \frac{y_1 R}{W} dx$$

$$\Rightarrow y_p(x) = -f(x) \int \frac{xf(x) \cdot f(x)}{f^2(x)} dx + xf(x) \int \frac{f(x) \cdot f(x)}{f^2(x)} dx$$

$$\Rightarrow y_p(x) = -\frac{x^2}{2}f(x) + x^2f(x) = \frac{x^2}{2}f(x)$$

and

$$y_c(x) = (ax + \beta)f(x)$$

$$\Rightarrow y(x) = y_c(x) + y_p(x)$$

$$\Rightarrow y(x) = \left(\frac{x^2}{2} + ax + \beta \right) f(x)$$

Hence, option (b) is correct.